IDENTITY & ACCESS MANAGEMENT

Authentication Log Analysis Report

Practical security review of Kerberos authentication events, account activity and audit-policy visibility.

Assessment type	IAM / Windows authentication monitoring
Environment	THM training domain (TRYHAT...THM as shown in evidence)
Evidence window	3 February 2026 (timestamps visible in Splunk evidence)
Report date	4 November 2026
Evidence source	Windows Security events and Splunk search results supplied for review

Executive summary

The supplied evidence demonstrates a normal Kerberos authentication sequence for the account **luke.sullivan**, alongside broader service-ticket activity and a separate account-creation event involving **nathan.brooks**. The Group Policy screenshot also shows Advanced Audit Policy categories as **Not configured**, which may limit the consistency and depth of security-event collection.

AREA	OBSERVATION	ASSESSMENT
Kerberos flow	TGT request (4768), service ticket (4769), then successful logon (4624) for luke.sullivan.	Observed sequence; not inherently malicious.
Account activity	Event 4720 records nathan.brooks created by adm-luke.sullivan.	Validate authorization and change record.
Audit policy	Advanced Audit Policy categories shown as Not configured.	Potential logging and detection gap.

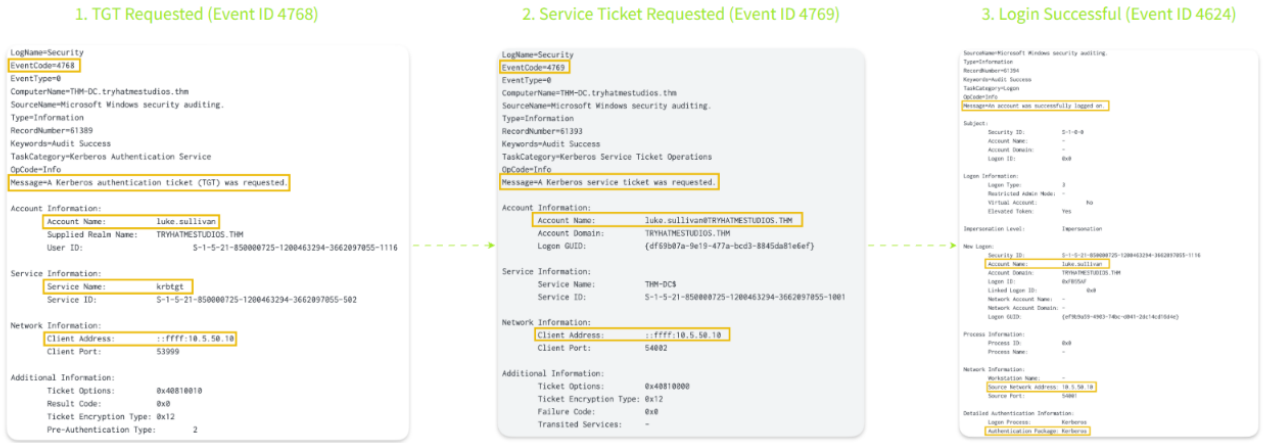
Overall conclusion: The evidence supports several investigation leads, but it does not by itself prove compromise or malicious activity. Confirm the business purpose, source host, account owner and approved change record before classifying any event as an incident.

1. Kerberos authentication sequence

The first screenshot correlates three Windows Security events associated with luke.sullivan and client address 10.5.50.10 (displayed in IPv4-mapped IPv6 form). Together, these events illustrate the expected Kerberos ticketing flow.

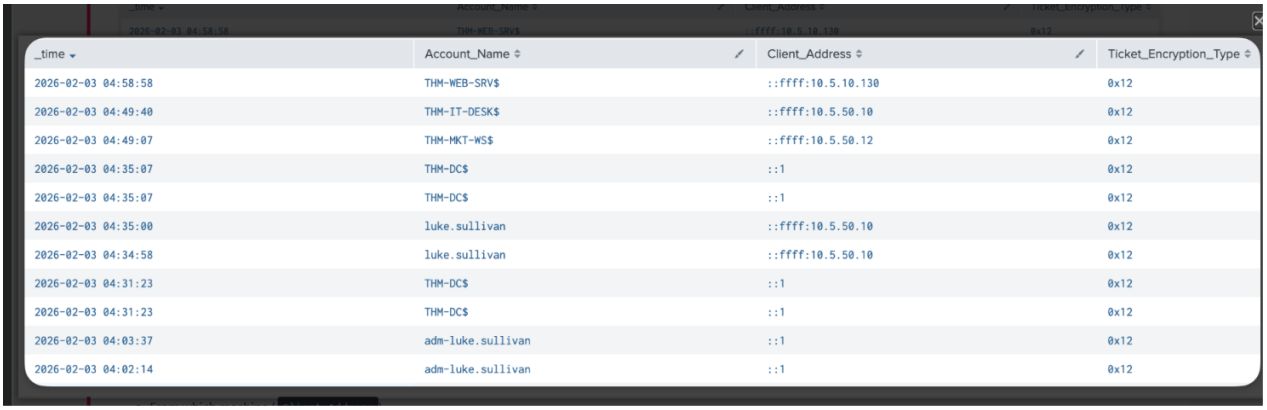
EVENT	MEANING	EVIDENCE INTERPRETATION
4768	Kerberos TGT requested	A Ticket-Granting Ticket request was recorded for luke.sullivan; service shown as krbtgt.
4769	Kerberos service ticket requested	A service ticket request was recorded for the domain controller service (THM-DC$).
4624	Successful logon	The logon event reports successful authentication using Kerberos.

Evidence 1 — Correlated Kerberos events



The sequence is consistent with a user obtaining a TGT, requesting a service ticket and then logging on. The sequence alone is not evidence of credential theft or lateral movement. Correlate it with the endpoint, user activity, expected working hours and subsequent resource access.

Evidence 2 — Kerberos service-ticket activity



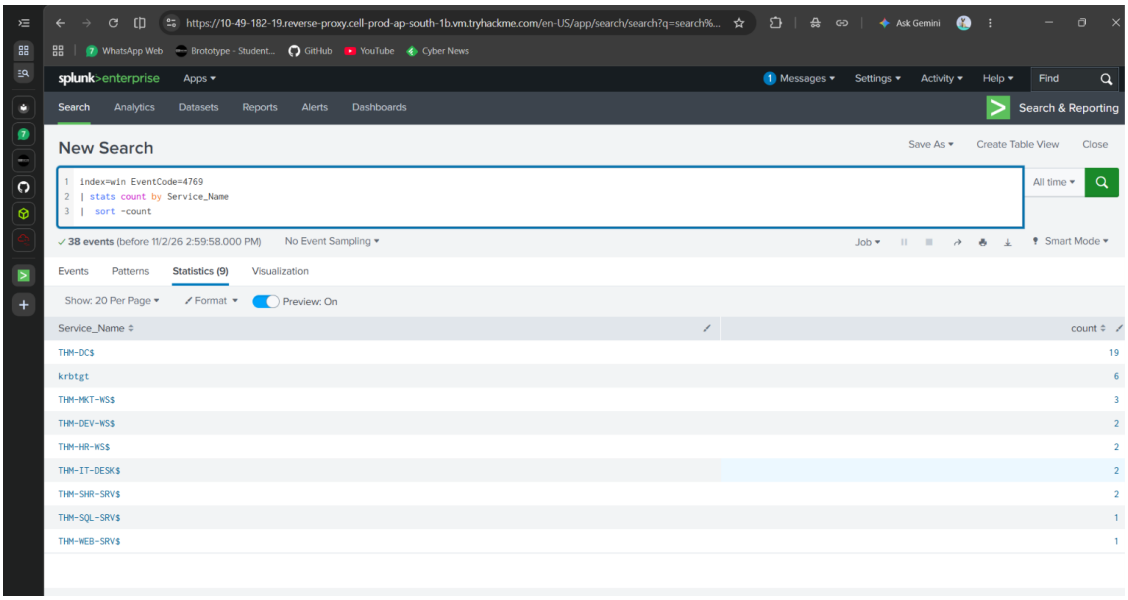
_time	Account_Name	Client_Address	Ticket_Encryption_Type
2026-02-03 04:58:58	THM-WEB-SRV$	::ffff:10.5.10.130	0x12
2026-02-03 04:49:40	THM-IT-DESK$	::ffff:10.5.50.10	0x12
2026-02-03 04:49:07	THM-MKT-WS$	::ffff:10.5.50.12	0x12
2026-02-03 04:35:07	THM-DC$	::1	0x12
2026-02-03 04:35:07	THM-DC$	::1	0x12
2026-02-03 04:35:00	luke.sullivan	::ffff:10.5.50.10	0x12
2026-02-03 04:34:58	luke.sullivan	::ffff:10.5.50.10	0x12
2026-02-03 04:31:23	THM-DC$	::1	0x12
2026-02-03 04:31:23	THM-DC$	::1	0x12
2026-02-03 04:03:37	adm-luke.sullivan	::1	0x12
2026-02-03 04:02:14	adm-luke.sullivan	::1	0x12

The supplied event list shows multiple accounts and service activity, with ticket encryption type 0x12 in the displayed rows. Encryption type 0x12 is AES256-CTS-HMAC-SHA1-96. Confirm the organisation's encryption baseline and investigate any unexpected legacy encryption types if they appear in the wider dataset.

2. Splunk investigation results

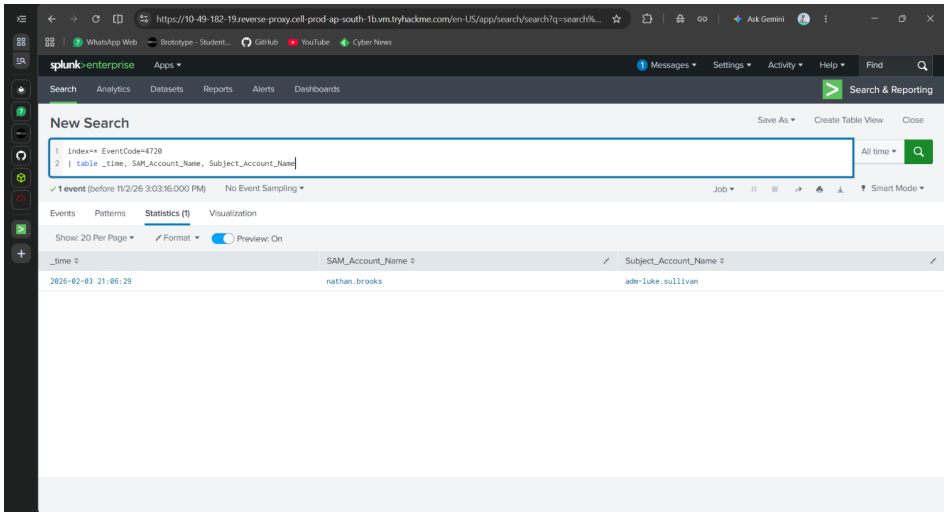
The Splunk searches use Windows event codes to summarize Kerberos service-ticket requests, review account creation and inspect TGT requests. The following observations are limited to the visible search results.

2.1 Service-ticket requests — Event ID 4769



The search returned 38 events. The aggregation shows THM-DC$ with 19 events, krbtgt with 6, THM-MKT-WS$ with 3, and several other service accounts with smaller counts. A high count is not automatically suspicious; compare it with a normal baseline and the expected role of each service.

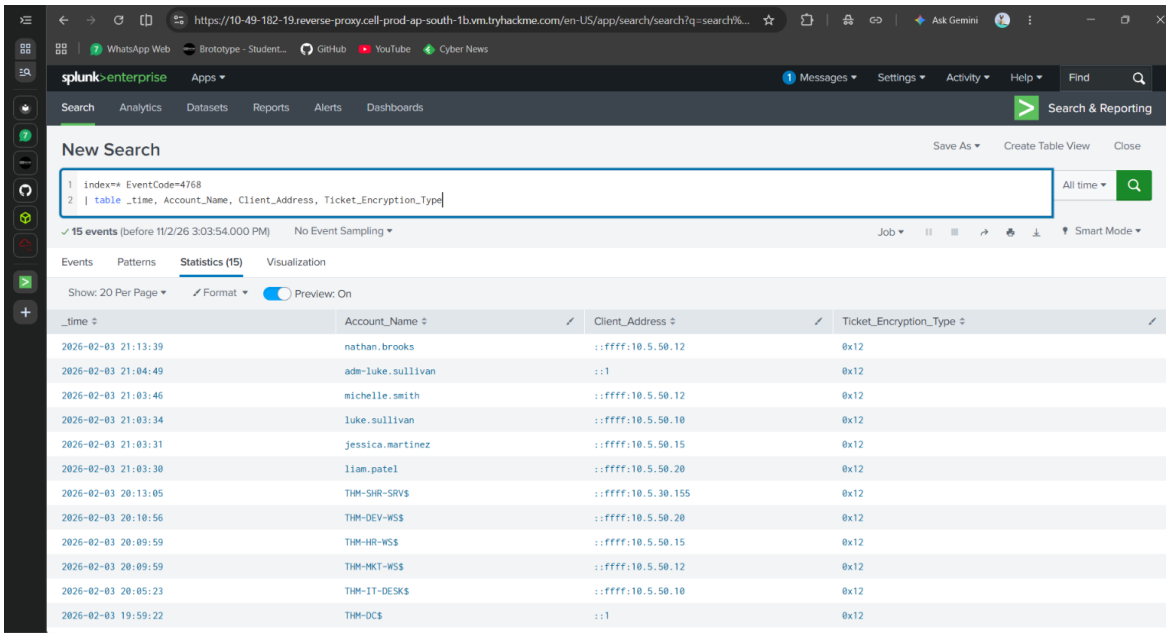
2.2 Account creation — Event ID 4720



The visible result records SAM_Account_Name **nathan.brooks** and Subject_Account_Name **adm-luke.sullivan** at 2026-02-03 21:06:29. This indicates that the named administrative account was the subject context associated with creation of the new account. Verify the request, approver, intended group memberships and whether the creation time aligns with an approved change.

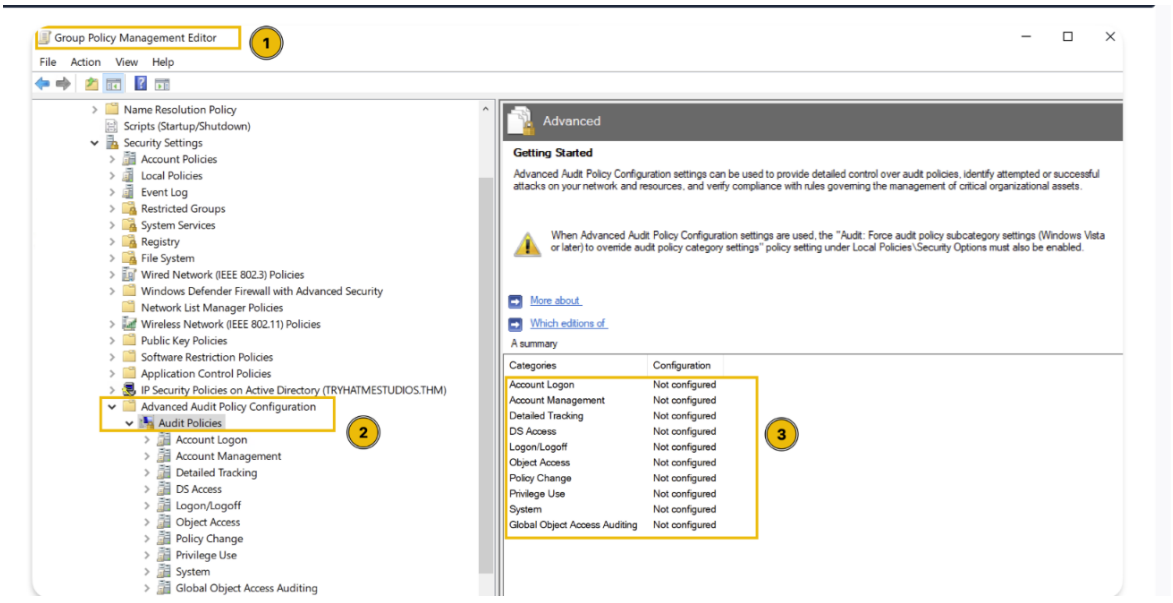
3. TGT review and audit-policy configuration

3.1 TGT requests — Event ID 4768



The search returned 15 events. The visible rows include standard user accounts, administrative accounts and computer accounts, with client addresses from the 10.5.50.0/24 lab network and loopback (::1). The displayed ticket encryption type is 0x12. Review administrative-account TGT requests separately and confirm that each source host and time is expected.

3.2 Advanced Audit Policy



The Group Policy Management Editor screenshot shows the listed Advanced Audit Policy categories as **Not configured**. This does not prove that no auditing is active: legacy audit policy, another linked GPO, local policy or domain-level configuration may apply. However, the screenshot indicates that this policy view does not explicitly define the displayed advanced audit subcategories.

4. Findings and recommendations

Findings are expressed as investigation priorities rather than confirmed incidents because the supplied screenshots do not include full event payloads, endpoint context or approved-change documentation.

ID / PRIORITY	FINDING	RECOMMENDED ACTION
F-01 **Medium**	New account nathan.brooks is shown as created under subject account adm-luke.sullivan.	Validate the ticket/change request, account owner, creation time, group membership and subsequent sign-ins. Disable or contain only if unauthorized activity is confirmed or policy requires it.
F-02 **Medium**	Advanced Audit Policy categories appear Not configured in the displayed GPO editor view.	Review effective policy using gpresult /h and domain-linked GPOs. Configure and test required audit subcategories, including Logon/Logoff, Account Management and Privilege Use, according to the organisation's baseline.
F-03 **Low–Medium**	Kerberos service-ticket and TGT activity is present, with multiple service accounts and client hosts.	Establish a baseline by account, service, host and time. Alert on unusual spikes, unexpected source hosts, privileged-account anomalies and ticket-encryption downgrade.

5. Suggested Splunk searches

Use the following as starting points; adjust field names and index/sourcetype to match the lab data.

Summarize 4769 requests by service

```
index=win EventCode=4769
| stats count by Service_Name
| sort - count
```

Review account creation events

```
index=* EventCode=4720
| table _time SAM_Account_Name Subject_Account_Name
```

Review Kerberos TGT requests

```
index=* EventCode=4768
| table _time Account_Name Client_Address Ticket_Encryption_Type
```

Conclusion

The exercise demonstrates how Windows authentication events and Splunk can be used to reconstruct Kerberos activity, identify account-management events and assess audit visibility. The key next steps are to validate the new-account creation, confirm effective audit-policy settings and establish baselines for Kerberos activity. No confirmed compromise can be concluded from the screenshots alone.

Scope note: This report is based solely on the six screenshots supplied. It is an educational/lab assessment, not a forensic certification or a statement about a production environment. Findings should be revalidated against original event records and organisational policy.

Appendix A — Failed logon investigation example

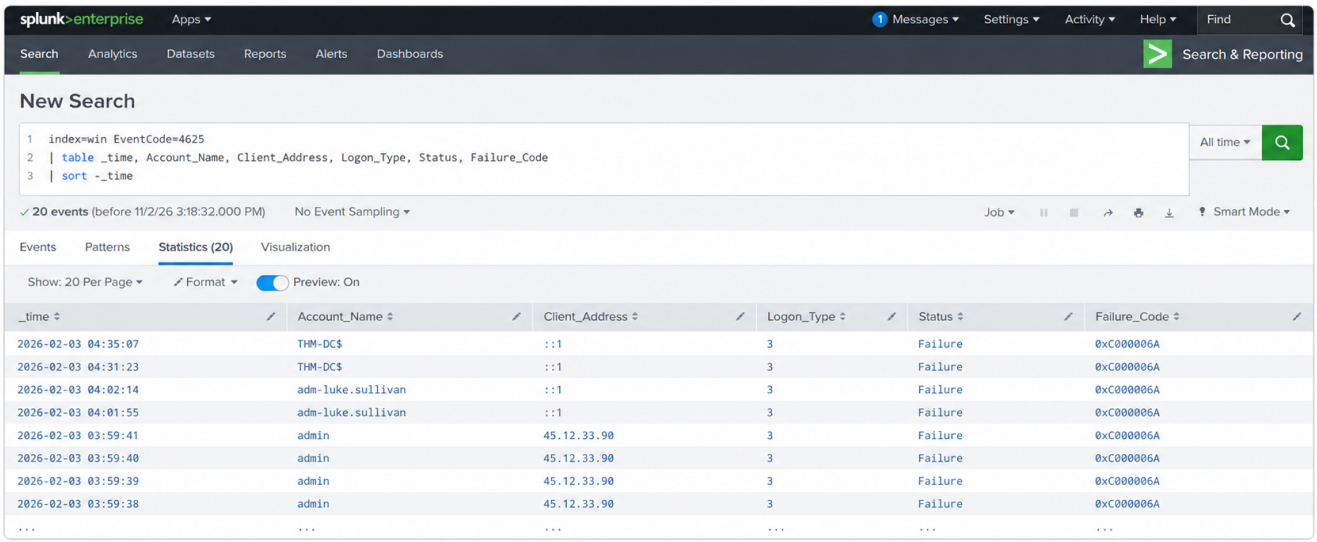
Event ID 4625 is used to investigate failed logon attempts. The example below illustrates a Splunk search and the type of fields an analyst may review.

ILLUSTRATIVE MOCK-UP — NOT A SCREENSHOT FROM YOUR LAB OR VERIFIED EVENT DATA. Replace with an actual Splunk capture before presenting this as completed hands-on evidence.

1.1 Failed Logon Attempts – Event ID 4625

The following Splunk query was executed to identify failed logon attempts:

```
index=win EventCode=4625
| table _time, Account_Name, Client_Address, Logon_Type, Status, Failure_Code
| sort -_time
```



Investigation notes: group failures by account and source address; check timestamps, logon type and failure status; then correlate with successful logons and endpoint context. A failure code such as 0xC000006A commonly indicates an incorrect password, but confirm the event details in the original Windows record.